

QUIZ - 2A  
STATISTICAL COMPUTING (MTH-210)

Name:

Roll Number

TIME 20 MINS.

MAXIMUM SCORE 15 MARKS

Please write your name and roll number, otherwise two marks will be deducted. This quiz has two questions and each question has two subparts. You have to write the correct answer only in the appropriate place. Keep all your codes with you till the end of this course. I might ask for it for cross verification. If you cannot provide the code then you will be assigned zero for that particular question.

1. Suppose  $(1.25, 2.25, 3.25)$  is a random sample from the following PDF

$$f(x; \mu) = 6(x - \mu)^2 e^{-2(x-\mu)^3}; \quad x > \mu.$$

You need to find the MLE of  $\mu$  using Newton-Raphson method without using any step-factor modification. Suppose  $\mu^{(0)} = 1.1$ , and you use stopping rule as  $|\mu^{(i)} - \mu^{(i+1)}| < 0.001$ , then if we denote  $A = \mu^{(1)}$  and  $B = \hat{\mu}$  [3 + 4 = 7]

**Answers:** The values of

A = 1.249752

B = 1.249505

2. You have to solve the same problem as in 1 using bisection method. Start the initial guesses as  $a^{(0)} = 1.00$  and  $b^{(0)} = 1.2499$  and stop the process when  $|b^{(i)} - a^{(i)}| < 0.001$ . If we denote  $A = \sqrt{a^{(1)} \times b^{(1)}}$  and  $B = (a^{(3)} + b^{(3)})/2.0$  [4 + 4 = 8]

**Answers:** The values of

A = 1.18578

B = 1.171806